Peter Kratz

pkratz22@gmail.com | 1-347-456-9431 | New York, NY

Work Experience

InformedDNA Remote
Data Engineer March 2025 - Present

- Designed and deployed production AWS serverless pipeline (Step Functions, Lambda, CDK) orchestrating claims processing, R-based report generation, and automated QuickSight dashboard refreshes, reducing client report turnaround from 15+ minutes of manual engineering effort to under 1 minute upon file receipt.
- Built multi-Lambda SAM application automating patient enrollment and eligibility synchronization between Formstack, S3, and Zoho CRM, processing 100K+ records daily with structured logging and error handling, eliminating 2 hours of manual work per week.
- Developed ETL pipelines standardizing 200+ claims files (pharmacy, medical, and professional) across 15+ client formats into unified schema, with automated S3 archival, deduplication, and data quality flagging, eliminating manual pre-processing and ensuring consistent downstream data availability.
- Created AWS data quality toolkit auditing Athena views for staleness, profiling query performance across 30+ databases and 300+ tables/views, and tracing cross-service references (QuickSight, Glue, Lambda) to safely decommission deprecated resources.

Coriell Life Sciences Philadelphia, PA
Senior Data Analyst March 2022 - March 2025

- Built composite pharmacogenomic risk scoring model weighting 6 clinical factors to rank 17K+ members into deciles across 5 groups, optimizing program enrollment targeting for 2 clients.
- Evaluated clinical program ROI by applying propensity score matching (MatchIt) to identify matched control groups, then running difference-in-difference and paired t-test models to compare healthcare costs and utilization across staggered treatment timings.
- Developed simulation-based cost savings projection tool sampling from 1M+ historical records with demographic stratification, enrollment ramp-up schedules, and savings decay modeling, enabling prospective clients to assess program viability for their population before committing to full claims data transfer.
- Built quarterly business reporting CLI with automated CRM/database merging, SQL generation, mismatch detection, and property-based testing to produce identified and deidentified cohort reports across 20+ client accounts.

Coriell Life Sciences Philadelphia, PA
Data Analyst October 2020 - March 2022

- Forecasted enrollment using changepoint detection (PELT) to isolate steady-state periods, then compared ETS, ARIMA, and Holt-Winters models with automatic quarterly fallback when monthly MAPE exceeded 30%, producing 12-month forecasts with confidence intervals.
- Automated monthly billing reconciliation across multiple clients by pulling CRM data via API and applying client-specific business rules, eliminating manual CRM lookups per client and reducing risk of invoicing errors.
- Designed outreach list generation pipelines combining S3 eligibility data with CRM enrollment status, applying household deduplication, minor filtering, and claims filtering to produce targeted mailing, email, and phone lists across 5+ clients on repeatable monthly cadence.

Education

Villanova University Villanova, PA
Bachelor of Business Administration Honors: *Cum Laude* May 2020
Majors: Business Analytics, Management Information Systems | Minors: Computer Science

Skills

Languages & Tools: Python (pandas, boto3, requests), R (tidyverse), SQL, Git
Analytics & Visualization: QuickSight, R (ggplot2, R Markdown)
AWS: Athena, CDK, Glue, Lambda, S3, SAM, Step Functions